

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech 3rd Semester Mid-Term Examination, September 2023

Subject: Mathematics-III(MA-2101)

(All Engineering Branches)

Time : 2 hours

Full Marks : 30

Answer any five questions

1. a) State axiomatic definition of probability. Using this definition prove that

b) $0 \leq P(B) \leq 1$, for any event B ,

c) $P(\bar{A} \cap B) = P(B) - P(A \cap B)$, for any two events A and B .

[2+2+2]

2. a) For n events $A_1, A_2, \dots, A_n$ connected to a random experiment E , show that

$$P(A_1 \cap A_2 \cap \dots \cap A_n) \geq \sum_{i=1}^n P(A_i) - (n-1).$$

b) If $A_1, A_2, \dots, A_n$ are mutually independent events with respective probabilities, $p_i = P(A_i) = 1 - \frac{1}{2^i}$; $i = 1, 2, \dots, n$. Prove that $P(A_1 \cup A_2 \cup \dots \cup A_n) = 1 - (\frac{1}{2})^{n(n+1)/2}$.

[3+3]

3. State and prove Baye's theorem.

[2+4]

4. a) Find the mean of the Binomial distribution and show that Poisson distribution is a limiting case of Binomial distribution.

b) A radio active source emits on the average 2.5 particles per seconds. Calculate the probability that 2 or more particles will be emitted in an interval of 4 seconds.

[(2+2)+2]

5. a) Find the constant k so that the function

$$f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

is a probability density function. Find (i) distribution function and (ii) $P(1 < X < 2)$.

b) A random variable X has the density function

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 1/2, & 1 < x \leq 2 \end{cases}$$

Find the mean.

[(1+2+1)+2]

P.T.O

6. Consider the set of equations:

$$4x_1 + 2x_2 - 3x_3 = 1, 6x_1 + 4x_2 - 5x_3 = 1$$

Verify that $x_1 = 2$, $x_2 = 1$, and $x_3 = 3$ is a feasible solution. Reduce this to a basic feasible solution. [1+5]

7. Show that $x_1 = 5$, $x_2 = 0$, and $x_3 = -1$ is a basic solution of the system of equations

$$x_1 + 2x_2 + x_3 = 4, 2x_1 + x_2 + 5x_3 = 5$$

Find the other basic solutions, if there be any.

[1+5]